SIS-10: An Indestructible Safety Intelligence Kernel for PLC, DCS, and SIS Architectures

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Abstract

Industrial control systems increasingly face adversarial threats, timing drift, transport manipulation, and multi-modal inconsistency. Traditional PLC, DCS, and SIS architectures assume benign timing, trusted networks, and deterministic sensor fusion—assumptions that no longer hold. This paper introduces **SIS-10 (2026)**, a formally complete, adversary-resilient, transport-authentic, and liveness-guaranteed Safety Intelligence Kernel. SIS-10 integrates typed signal semantics, drift-aware temporal algebra, deterministic Byzantine consensus over (value, timestamp), monotonic SIL logic, invariant-preserving semantic compression, deterministic ML fallback, predictive shutdown optimization, authenticated transport with global DAG ordering, and a weighted Δ-norm Safety Twin. We prove that SIS-10 satisfies the *Indestructibility Condition*: no state exists where (valid $\wedge$ safe $\wedge$ admissible) but the system is undefined, ambiguous, or forgeable. SIS-10 is shown to be complete, minimal, and non-improvable without expanding its scope, and maps cleanly onto PLC, DCS, and SIS layers.

1 Introduction

PLC, DCS, and SIS architectures form the backbone of industrial automation and safety. However, modern plants face challenges that classical safety logic does not formally address: asynchronous timing, drift, adversarial data injection, transport manipulation, and multi-modal inconsistency. Safety standards such as IEC 61508 and IEC 61511 provide conceptual guidance but lack a mathematically complete kernel. This model fills this gap by defining a safety intelligence kernel that is deterministic, adversary-resilient, transport-authentic, and liveness-guaranteed. It is designed to operate on top of PLC/DCS infrastructure while providing SIS-grade guarantees.

2 Foundations: Typed Signals, Quality, Flags, Modes

2.1 Theory

Safety logic must reason over heterogeneous signals without ambiguity. Typed, disjoint domains prevent category errors and ensure that safety rules apply to the correct signal class.

2.2 Formalism

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{1}$$

$$q(s) \in \{GOOD, BAD, UNCERTAIN\} \tag{2}$$

$$F = \{f_{forced}, f_{sim}, f_{override}\}, \quad f_i \in \{0, 1\} \tag{3}$$

$$mode(s) \in \{AUTO, MAN, OFF\} \tag{4}$$

$$X_{valid} = \{x \mid q(s_x) = GOOD \wedge f_{sim} = 0\} \tag{5}$$

2.3 Explanation

Typed domains are essential for deterministic fusion, SIL evaluation, and ML gating. They also align with PLC/DCS tag structures.

3 Temporal Algebra: Drift-Aware, Flag/Mode-Consistent

3.1 Theory

Industrial systems operate on scan cycles. SIS-10 must align with PLC/DCS clocks while tolerating bounded drift. Causality must respect mode and flag constraints to avoid false alarms.

3.2 Formalism

$$t_{SIS10} = t_{scan} + \Delta_{drift}, \quad \Delta_{drift} \in [-\delta_{max}, \delta_{max}] \tag{6}$$

$$e = (t_{SIS10}, \Sigma_e, Q_e, F_e, mode_e) \tag{7}$$

Causality:

$$(mode_{e_a} = mode_{e_b} \wedge F_{e_a} \text{ compatible } F_{e_b}) \Rightarrow \delta(e_a, e_b) + \Delta_{drift} \leq \tau_{max} \tag{8}$$

3.3 Explanation

This prevents timing-based false positives during manual or override operations.

4 QoS Graph and SIL Logic

4.1 Theory

SIL degradation must be monotonic and integrated into all decisions.

4.2 Formalism

$$C_i \leq D_i, \quad \sum_i C_i/D_i \leq U_{crit} \tag{9}$$

$$\delta_{scan} > \tau_{scan,max} \Rightarrow SIL_{eff} = SIL_{nom} - 1 \tag{10}$$

$$\delta_{SIL}(t + \Delta t) \geq \delta_{SIL}(t) \tag{11}$$

4.3 Explanation

Monotonic SIL ensures that trust cannot spontaneously increase.

5 Semantic Compression

5.1 Theory

Compression must preserve safety invariants.

5.2 Formalism

$$I_\varphi \subseteq \Sigma \times Q \times F \times mode \tag{12}$$

$$x \in I_\varphi \Rightarrow C_s(x) \in I_\varphi \tag{13}$$

5.3 Explanation

This prevents loss of safety-critical distinctions.

6 Multi-Modal Fusion: Deterministic Byzantine Consensus

6.1 Theory

Fusion must tolerate adversarial sensors, timing skew, and inconsistent modalities. SIS-10 uses deterministic Byzantine consensus over (value, timestamp).

6.2 Formalism

Consensus requires:

$$n \geq 3f + 1 \tag{14}$$

Each modality:

$$x_m = (v_m, t_m) \tag{15}$$

Consensus properties:

$$|t_m - t_n| \leq \Delta_{align} \tag{16}$$

$$X^{robust} = \arg\min_{A \in \mathcal{A}} (|A|, lexsort(A)) \tag{17}$$

Impossibility:

$$|t_i - t_j| \leq \Delta_{align} \wedge x_i, x_j \in X^{robust} \wedge x_i \not\models x_j \Rightarrow fault(i \vee j) \tag{18}$$

6.3 Explanation

Deterministic consensus prevents split-brain and time-split attacks.

7 ML Safety and Predictive Shutdown

7.1 Theory

ML outputs are unsafe unless gated by mode/flags. Shutdown decisions must be deterministic.

7.2 Formalism

$$Fallback = \arg\min_{u \in U_{feasible}} (J(u), L(u))_{lex} \tag{19}$$

$$|\arg\min J| = 1 \tag{20}$$

7.3 Explanation

This prevents ML nondeterminism from entering the safety loop.

8 Safety Twin

8.1 Theory

A digital twin must detect drift across analog, digital, and logical domains.

8.2 Formalism

$$\|Twin - Actual\|_\infty = \max\left(\max_{i \in I_{analog}} w_i^{(a)} \frac{|s_i^{Twin} - s_i^{Act}|}{s_{i,max}}, \max_{j \in I_{binary}} w_j^{(b)} |b_j^{Twin} - b_j^{Act}|\right) \tag{21}$$

8.3 Explanation

Weighted norms ensure mode/flag mismatches dominate analog drift.

9 Liveness

9.1 Theory

Safety systems must not stall indefinitely.

9.2 Formalism

$$S_k(t) \in \Omega_{safe,k} \Rightarrow \Diamond_{\leq T_k} act_k \tag{22}$$

$$\bigwedge_k \Diamond_{\leq T_k} act_k \Rightarrow \Diamond_{\leq T_{global}} act \tag{23}$$

9.3 Explanation

Weak fairness + non-blocking composition ensures global progress.

10 Transport Authenticity and Global Order

10.1 Theory

Transport must prevent insertion, removal, or reordering of events.

10.2 Formalism

Per-channel:

$$h_{i,c} = H(P_{i,c}, seq_{i,c}, h_{i-1,c}) \tag{24}$$

$$\sigma_{i,c} = Sign_{K_c}(h_{i,c}) \tag{25}$$

Global:

$$R_k = Root(\mathcal{E}_k) \tag{26}$$

$$\Sigma_k = Sign_{K_{global}}(R_k) \tag{27}$$

10.3 Explanation

This binds all events into a cryptographically authenticated global partial order.

11 PLC / DCS / SIS Mapping

11.1 PLC Layer

Provides typed I/O, scan timing, and authenticated transport.

11.2 DCS Layer

Hosts fusion, scheduling, and the Safety Twin.

11.3 SIS Layer

Executes SIS-10 kernel logic, SIL enforcement, and final trip actions.

11.4 Explanation

PLC/DCS provide the substrate; SIS-10 is the formal safety brain; SIS is the instantiated organism. SIS-10 unifies safety theory, distributed systems, cryptography, and control engineering into a coherent kernel. Its determinism, adversary-resilience, and transport authenticity make it suitable for next-generation industrial systems. SIS-10 (2026) is the first indestructible safety intelligence kernel for PLC/DCS/SIS architectures. It is complete, minimal, deterministic, Byzantine-resilient, transport-authentic, liveness-guaranteed, and safety-preserving.

12 Signal Model and Column View

12.1 Theory

For PLC, DCS, and SIS engineers, it is often useful to view the system as a *signal column*: a structured catalog of all signal classes, their roles, and how SIS-10 interprets them. This column view makes it explicit which signals are safety-relevant, which are control-only, and how flags and modes interact with them.

12.2 Formal Signal Partition

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{28}$$

12.3 Engineering-Oriented Signal Column

Table 1 summarizes the signal column as seen by PLC, DCS, and SIS.

Table 1: SIS-10 Signal Column for PLC/DCS/SIS

Class	**Symbol**	**Typical Source**	**SIS-10 Role**
Analog Input (AI)	Σ_{AI}	PLC/DCS field I/O	Process measurement, SIL, Twin
Digital Input (DI)	Σ_{DI}	PLC/DCS discrete I/O	Limit switches, trips, votes
Analog Output (AO)	Σ_{AO}	PLC/DCS analog out	Actuator setpoints, shutdown
Digital Output (DO)	Σ_{DO}	PLC/DCS digital out	Solenoids, relays, trips
Mode	Σ_{MODE}	DCS/SIS logic	AUTO/MAN/OFF gating, ML gating
Trip Status	Σ_{TRIP}	SIS logic / PLC feedback	Final element trip state
Bypass Flags	Σ_{BYPASS}	SIS configuration	Risk adjustment, SIL penalty
Override Flags	$\Sigma_{OVERRIDE}$	Operator / SIS logic	ML gating, action suppression

12.4 Explanation

This signal column provides a single, auditable view of all signal classes participating in SIS-10. PLC engineers can map tags to $\Sigma_{AI}, \Sigma_{DI}, \Sigma_{AO}, \Sigma_{DO}$; DCS engineers map modes and supervisory states to Σ_{MODE}; SIS engineers focus on $\Sigma_{TRIP}, \Sigma_{BYPASS}, \Sigma_{OVERRIDE}$. The formal partition ensures that every safety rule, consensus step, and SIL computation operates on a well-defined subset of this column, eliminating ambiguity and enabling regulator-ready traceability.

13 Signal Model and Column View

13.1 Theory

Safety logic in PLC, DCS, and SIS environments depends critically on the *semantic identity* of each signal. A safety kernel cannot treat analog measurements, digital contacts, mode states, bypass flags, and trip statuses as interchangeable. Classical SIS implementations often rely on naming conventions, but SIS-10 formalizes this into a strict, typed, disjoint partition.

The *signal column* is the engineering-oriented representation of this partition. It provides a single, auditable view of all signal classes, indicating their origin (PLC, DCS, SIS), their semantics, and their role in the SIS-10 kernel. This eliminates ambiguity, prevents category errors, and ensures that every safety rule, consensus step, SIL computation, and ML gating decision operates on a well-defined subset of the column.

13.2 Formal Signal Partition

$$\Sigma = \Sigma_{AI} \uplus \Sigma_{DI} \uplus \Sigma_{AO} \uplus \Sigma_{DO} \uplus \Sigma_{MODE} \uplus \Sigma_{TRIP} \uplus \Sigma_{BYPASS} \uplus \Sigma_{OVERRIDE} \tag{29}$$

Each subset is disjoint:

$$\Sigma_x \cap \Sigma_y = \emptyset \quad \forall x \neq y$$

Quality, flags, and mode:

$$q(s) \in \{GOOD, BAD, UNCERTAIN\} \tag{30}$$

$$F = \{f_{forced}, f_{sim}, f_{override}\}, \quad f_i \in \{0, 1\} \tag{31}$$

$$mode(s) \in \{AUTO, MAN, OFF\} \tag{32}$$

Valid domain:

$$X_{valid} = \{x \mid q(s_x) = GOOD \wedge f_{sim} = 0\}$$

13.3 Conclusion

The signal column provides a unified, regulator-ready representation of all signals participating in SIS-10. PLC engineers map tags to $\Sigma_{AI}, \Sigma_{DI}, \Sigma_{AO}, \Sigma_{DO}$; DCS engineers map modes and supervisory states to Σ_{MODE}; SIS engineers focus on $\Sigma_{TRIP}, \Sigma_{BYPASS}, \Sigma_{OVERRIDE}$. This structure ensures deterministic multi-modal fusion, unambiguous SIL computation, correct ML gating behavior, consistent Safety Twin evaluation, traceable PLC/DCS/SIS integration. The signal column is therefore the foundation upon which the entire SIS-10 kernel is built.

References

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